

Project Report: AI Language Translation Tool

Project Name: Language Translation Tool

Technology Used: Python, Streamlit, Google Translate API

Project Type: Artificial Intelligence Based Mini Project

1. Project Description

The Language Translation Tool is an Artificial Intelligence based application that allows users to translate text from one language to another using a translation API. The project provides a simple user interface where users can enter text, choose source and target languages, and receive translated output instantly.

2. Objectives of the Project

- To create a user-friendly language translation interface.
- To integrate translation APIs for accurate language conversion.
- To display translated text clearly on the screen.
- To improve communication between users of different languages.
- To understand API integration and AI-based applications.

3. Features of the Project

- Translate text between multiple languages
- Simple and interactive user interface
- Fast translation response
- Copy translated text feature
- Text-to-speech support (optional)
- Supports real-time translation

4. Technologies Used

Technology	Purpose
Python	Backend Programming Language
Streamlit	User Interface Development

Google Translate API	Language Translation
VS Code	Code Editor
GitHub	Project Hosting and Version Control

5. Project Workflow

1. User enters text in the input box.
2. User selects source and target languages.
3. Application sends text to translation API.
4. API processes the translation request.
5. Translated text is returned and displayed to the user.

6. Folder Structure

```

Language_Translator/
|
├── app.py
├── requirements.txt
├── README.md
├── screenshots/
|   ├── input_screen.png
|   └── output_screen.png

```

7. Sample Python Code

```
main.py
1  import sys
2  import streamlit as st
3  from translate import Translator
4
5  #Language mapping
6  language_mapping = dict(english="en", french="fr", spanish="es", italian="it", German="de", Chinese="zh", Japanese="jp",
7                          Portuguese="pt", Hindi="hi", Bengali="bn")
8
9  #streamlit app title
10 st.title("Language Translator")
11
12 #user input for text and language selection
13 text_to_translate = st.text_area("Enter text to translate:")
14 target_language = st.selectbox(
15     label="Enter target language:", list(Language_mapping.keys()))
16
17 # Translate Button
18 if st.button("translate"):
19     if text_to_translate :
20         target_language = language_mapping[selected_language_name]
21         try:
22             #Initialize the translator with the target language
23             translator = Translator(to_lang=target_language)
24             # perform the translation
25             translated_text = translator.translate(text_to_translate)
26             # Display the translated text
27             st.write[ "app: |" + translated_text + " ", translated_text ]
28         except Exception as e:
29             st.error(f"Translation failed: {e}")
30     else:
31         st.warning("Please enter text to translate")
```

8. Project Screenshot

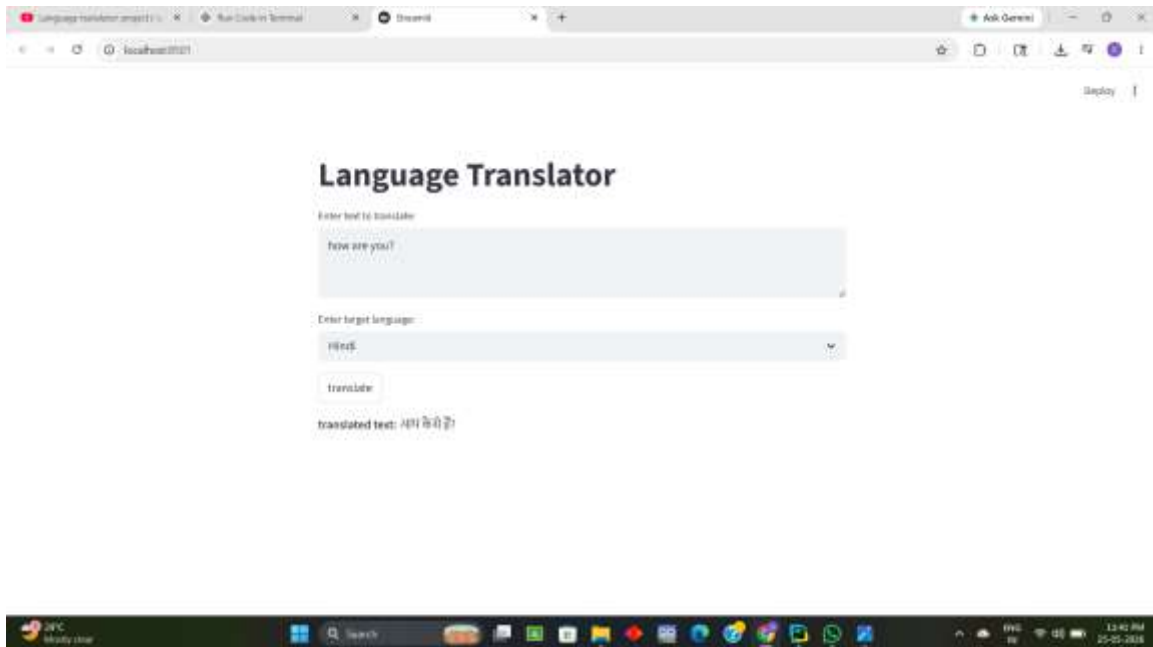
✓

TASK 1: Language Translation Tool

- **Create a user interface** where user can enter text and select source & target languages.
- **Use a translation API** like Google Translate API or Microsoft Translator to process the input.
- **Send the text** to the API and get the translated response.
- **Display the translated text** clearly on the screen.
- **Optional:** Add a copy button or text-to-speech feature for better usability.

Figure: Task description screenshot of Language Translation Tool project.

9. Output Screenshot



11. Advantages of the Project

- Easy communication between different language users
- Fast and accurate translation
- Simple user interface
- Useful for students and professionals
- Demonstrates AI and API integration

12. Future Improvements

- Add voice input support
- Add speech output feature
- Support offline translation
- Improve UI design
- Add document translation support

13. Conclusion

The Language Translation Tool successfully demonstrates the use of Artificial Intelligence and APIs to translate text between languages. The project is easy to use, efficient, and helps users communicate effectively across different languages.